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Question 1

Answer saved

Points out of 10.00

Soft tissue follows an exponential deformation behavior in uniaxial tension while it is in the physiologic or normal range of elongation. This can be expressed as

$$\sigma = \frac{E_0}{a}(e^{a\epsilon} - 1)$$

where σ = stress, ϵ = strain, and E_0 and a are material constants that are determined experimentally. To evaluate the two material constants, the above equation is differentiated with respect to ϵ , which is a fundamental relationship for soft tissue

$$\frac{d\sigma}{d\epsilon} = E_0 + a\sigma$$

To evaluate E_0 and a , stress-strain data are used to plot $d\sigma/d\epsilon$ versus σ and the slope and intercept of this plot are the two material constants, respectively. The table contains stress-strain data for heart chordae tendineae (small tendons use to hold heart valves closed during contraction of the heart muscle). This is data from loading the tissue; different curves are produced on unloading.

$\sigma \times 10^3 \text{ N/m}^2$	87.8	96.6	176	263	351	571	834	1229	1624	2107	2678	3380	4258
$\epsilon \times 10^{-3} \text{ m/m}$	153	204	255	306	357	408	459	510	561	612	663	714	765

Match the σ and ϵ values to the corresponding

$$\frac{d\sigma}{d\epsilon}$$

values using finite differencing based on the given data. You can calculate all the values first and then do the matching.

1. $\sigma = 87.8$ and $\epsilon = 153$
2. $\sigma = 2677$ and $\epsilon = 663$
3. $\sigma = 350$ and $\epsilon = 357$
4. $\sigma = 1227$ and $\epsilon = 510$
5. $\sigma = 96.6$ and $\epsilon = 204$
6. $\sigma = 176$ and $\epsilon = 255$

0.1725	1
0.8647	5
1.6314	6
3.0196	3
7.7451	4
12.4804	2

◀ Quiz 2

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Quiz 4 ▶

